

Analysis of Geologic Data Final Project

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General Instructions

As many of you have discovered over the past several months having code and statistics taught to you and actually trying to do it yourself are two very different experiences. Further, I've primarily been giving you datasets to work with which are already pre-processed to some degree (i.e. variables removed, data transformed, missing values filled in, etc.). For the final project you will be working with unsterilized raw data. By doing so you will have to make decisions about how to incorporate, transform, or cut out those anomalies in your analyses and consider what biases those choices might introduce.

The final project will be a written report (with maps, graphs, and tables where appropriate) with three parts. **1)** An introduction to the dataset and question being investigated, **2)** a methods section explaining the steps involved in statistical analyses, including assumptions, exceptions, or transformations of the data, and **3)** a results section detailing the results of your analyses. Because this is not a full-on research project you are not expected to have the background knowledge and context to write a more standard research paper with a discussion section on the implications of your findings. Your goal here is to demonstrate competence in writing your own code to perform statistical analyses in the Python programming environment.

Along with the written report you will need to submit a Python Jupyter Notebook containing all the steps required for your analysis. This means the code must also be commented adequately for someone else to understand what the code is doing.

The length of the paper is much less important than making sure your analysis is clearly and completely explained. However, including figures/tables, I expect the final reports will be 5+ pages (12-point font, single line spacing).

There will be several checkpoints over the remainder of the semester where you will receive feedback on draft versions before the final report is due on May 12th through UBLearn as show below.

Checkpoint	Due Date	Project Grade	Outcome
Pick topic (propose bonus analysis)	April 2, 11:59PM	5%	Approval

Checkpoint	Due Date	Project Grade	Outcome
Draft Section 1	April 28, 11:59PM	15%	Draft comments
Presentation (grads only)	May 5 Class	15%	Awesome Presentations
Project Due	May 12, 11:59PM	80%/65%	Final Project Grade

Submission requirement

1. Pick Topic: Submit your selected topic in a word document to UBLearn. If you plan to use one of the example dataset, you would only need to include the dataset title in the word document. If you plan to use your own dataset, you will need to provide the dataset and your proposed analysis method. This proposed method would have four bullet points as shown in the **Example datasets and corresponding data analysis**.
2. Draft Section 1: Submit all necessary documents to UBLearn (More details can be referred to the section **Checkpoint 1 Partial Draft** below)

Graduate Students

Graduate students may propose their own datasets and analyses but they need to be approved by me (the instructor). Otherwise they are responsible for the written final project as outlined above. In addition, graduate students will give a ~15 minute presentation during the last week of classes (April 30th or May 2nd). These should be in the style of a conference talk.

Undergraduate Students

Undergraduate students must choose one of the three datasets below for their final project. The source, basic information, and goals of each dataset are listed in more detail below. The actual dataset have also been posted if you wish to look at them before choosing (**warning**: some of the datasets are quite large and will take more than a few seconds to download).

Example datasets and corresponding data analysis

Dataset 1) Global distribution and geochemistry of Neogene-Quaternary intraplate volcanic rocks

Source: <https://data.goettingen-research-online.de/dataset.xhtml?persistentId=doi:10.25625/R3BRK>

Variables

Latitude, longitude, geographic location, rock type, alkalinity, age estimate, uncertainty in age estimate, method of age estimate, geochemical composition

Goals of Data Analysis

- 1) create a map showing the distribution of volcanic rocks color-coded or symbolized by one of the other variables (ex. age).
- 2) exploratory analysis of variation in the dataset (ex. variability in alkalinity by rock type or age)

- 3) perform a PCA analysis on the samples
- 4) perform some kind of uncertainty or sensitivity analysis on the PCA results (ex. jackknife, bootstrap, permutation)

Dataset 2) Groundwater levels of arsenic and uranium in private well from Connecticut

Sources: <https://data.usgs.gov/datacatalog/data/USGS:58458f8de4b04fc80e5233c1>

Variables

Arsenic levels, uranium levels, bedrock unit, rock grouping, percent areal coverage of bedrock units in the state, areal coverage of bedrock coverage, age of bedrock units, shapefile of bedrock units

Goals of Data Analysis

- 1) Create a bedrock unit map of both arsenic and uranium levels
- 2) exploratory analysis of variation in the dataset (ex. correlation between uranium and arsenic levels)
- 3) test whether uranium levels are different among rock grouping and bedrock age
- 4) perform some kind of uncertainty or sensitivity analysis around your test from step 3

Dataset 3) Lake Powell physical and ecological data from 1964-2021

Source: <https://www.sciencebase.gov/catalog/item/62b39805d34e8f4977cb788c>

Variables

Latitude, longitude, times of sampling, temperature with depth, secchi depth, lake chemistry, phytoplankton survey, zooplankton survey

Goals of Data Analysis

- 1) Create a time series plot of temperature and one other variable of your choice through time, including a smoothed average
- 2) exploratory analysis of variation in the dataset (ex. skewness of values, pie charts, etc.)
- 3) perform a loess analysis on the time series and try to identify any periodic signals
- 4) perform a parametric test of the correlation between two variables in the dataset

Bonus points

For both undergraduate and graduate students, you will have the opportunity to get bonus points for your final project.

The maximum amount of the bonus points is 5% of the entire course grade (not 5% of the final project grade!). Note: that the final project account for 25% of the entire course grade. If you opt for bonus points, the maximum amount of the grade for final project is 30%.

How you can achieve bonus points:

1. Undergraduate students only: If an undergraduate student opt to conduct an oral presentation, you can receive a bonus $2.5\% \times \text{your presentation grade}$ (If you get 80% of your presentation, you will add 2% to your final grades). You will need to indicate whether you want to conduct an oral presentation in the "Pick Topic" document.
2. For both graduate and undergraduate students
 - a. For those of you using the provided dataset, propose one or two additional analysis and corresponding methodology in the "Pick Topic" document, and provide your justification. I will evaluate the methodology and provide comments and suggestions. One additional analysis that I approve would receive 2.5% cap increase.
 - b. For those of you using your own dataset, in addition to your originally proposed analysis (four bullet points), you would need to provide one or two additional analysis (so 5-6 bullet points in total). I will evaluate the methodology and provide comments and suggestions. One additional analysis that I approve would receive 2.5% cap increase.

Checkpoint 1 Partial Draft (15% of total)

Due: Submission through UBLearn by April 28th at 11:59PM with both a written report and a Python file containing code.

Draft Completion Criteria: For the datasets presented here goals 1 and 2 should be complete (i.e. base data visualization and variance description). For the submitted R file this means that your code for goals 1 and 2 should be fully executable and commented. For the text this means an you should have a full draft of the introduction section of your report and methods section which is complete for the operations in goals 1 and 2 (which may include descriptions of how missing/inapplicable data were handled, steps checking that certain statistical conditions were being met, etc.).

For those using other datasets which were approved the completed draft should follow a similar pattern. A complete base data visualization and variance description of your data fully commented as an Jupyter Notebook and a written report with a complete introduction and methods section completely covering the material in your Jupyter Notebook.

Generated figures should have axes labeled appropriately (i.e. no raw\$x as the x-axis label), be given a figure legend (see example below), and referenced in the text where appropriate (ex. “as depth increases oxygen levels rapidly decline (Fig. 4)..”).

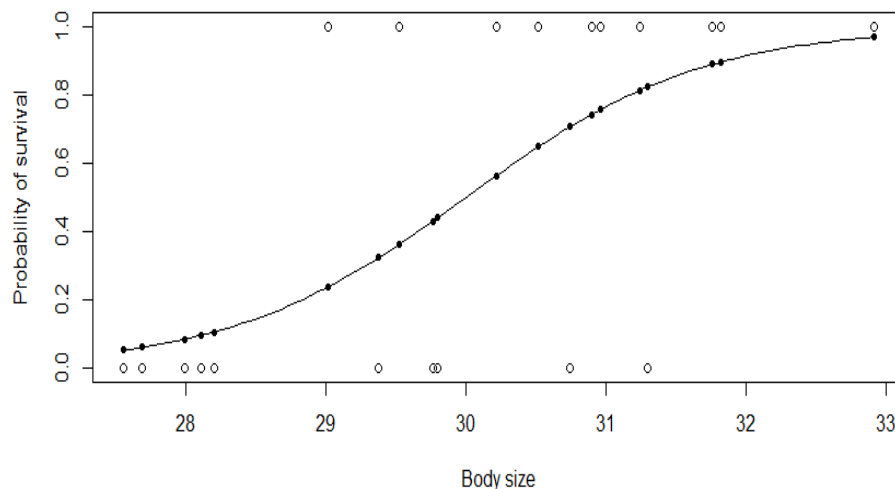


Figure 1: Logistic regression of the probability of species survival past the stage in which they originate dependent on their body size.

Rubric [15 total]**Python Jupyter Notebook File [6 points]**

- 1** – Minimal code or comments (i.e. mostly missing, not fully executable)
- 2** – Only partially complete (i.e. not fully executable) code for either goal and insufficient comments
- 3** – Complete code only for one of the two goals, and insufficient comments
- 4** – Complete code for both goals, but insufficient comments for both
- 5** – Complete code for both goals, but insufficient comments for one of the goals
- 6** – Complete code and comments for both goals

Written Report [6 points]

- 1** – Minimal writing for either introduction or methods
- 2** – Both sections partially complete (i.e. missing methods steps or incomplete description of dataset)
- 3** – One section completely missing, while other is mostly complete
- 4** – One section complete, other section partially complete
- 5** – Both sections almost complete, but missing some critical element or step (ex. forgot to include mention something that appears in your code)
- 6** – Both introduction and relevant methods sections complete (ex. all code steps mentioned)

Grammar & Spelling [3 points]

- 1** – errors in grammar or spelling are common and/or the writing is very difficult to understand
- 2** – errors in grammar or spelling are uncommon but persistent, some areas are difficult to understand
- 3** – errors in grammar or spelling are rare, no difficulties in understanding writing

Presentation [15% total, graduate students required; undergrad bonus points]

Length [2 points]

12-15 minutes (conference style)

Format

A powerpoint, Google Slides, etc. presentation, **to be submitted on the day you present to Dr. Cheng via email.**

Figures

Every slide can have a figure if that makes sense for your presentation. I'd much rather you have figures and explain them well than a wall of text you are reading from. Every figure should have axes labeled and a legend if appropriate.

Contents [8 points total]

Your presentation should essentially be an oral version of your written report. Specifically, it should contain at least the following elements. They do not have to be laid out in the linear order below, but all of them need to be included somehow in your presentation.

Introduction [1 point] – what dataset are you working with, including details about the form of the data

Methods [4 points] – bulk of your presentation here, focus on the data management decisions you made to import the data, what was excluded and why, any adjustments you had to do to make certain approaches/methods work, etc. Information on specific functions you used that were not introduced in class is fine, but it should not just be screenshots of your code.

Results [2 points] – summary tables and figures demonstrating the structure of your data and the outcomes of the statistical methods you described in the methods section

Conclusion [1 point] – summary of the major methodological challenges you faced in analyzing the dataset and what you might have done differently knowing what you do now

Delivery [5 points]

- 1** – Presentation filled with spelling/grammar errors, text contradictory to what the speaker says, message very unclear
- 2** – Spelling/grammar errors persistent in the presentation, portions where message is very unclear
- 3** – Spelling/grammar errors are minor, one portion of the presentation is poorly explained
- 4** – Spelling/grammar errors are non-existent/very minor, presentation is good but goes significantly over or under time
- 5** – Spelling/grammar errors are non-existent/very minor, presentation is clear and falls in the timed window